

Regression for Yes/No Outcomes: Logit

MKT 566 · Fall 2026

Davide Proserpio

USC Marshall School of Business

Before We Start

Housekeeping

- **Homework 2** (box office, OLS and logit) is out, due **Thursday, Oct. 8**. Everything it needs is in Thursday's deck and today's
- **Group project**: proposal presentations on **Oct. 13 and 15**
- Today: **logit** for about 40 minutes, then the **Airbnb case** on your laptop (aim to finish Task 5 in class; report due **Sunday, Sept. 27**)
- Thursday: **clustering**

From “how much” to “whether”

Thursday’s regressions predicted a **number**: sales, price. Many marketing questions are **yes or no**:

- Will this customer **buy**? **Churn**? **Click** the ad? **Open** the email?
- Will this loan **default**? Will this listing be a **hit**?

Today you learn to

1. See why a straight line is the wrong tool for a 0/1 outcome
2. Read a **logit** model: sign, odds, and predicted probabilities
3. Run it in R with `glm()` and put it in a **stargazer** table

What stays the same

- Predictors, dummies, base levels, stars, “holding the others fixed”: all as in OLS
- The AI writes the code. You choose the model and read the result

R script for today

In the [code/](#) folder on the course website, next to Thursday's:

1. [w4-2-logit-class.R](#): every chart and model in today's deck
2. [data/airbnb.csv](#): the same Airbnb listings

[w4-code.zip](#) has all of week 4: Thursday's two scripts, today's, and the data. If you downloaded it on Thursday, you already have everything.

A Yes/No Outcome

Our outcome: is a listing a “gem”?

An investor wants listings that are **both** well rated **and** actually booked. Define a **gem** as a listing with a rating of **4.5 or more** and **more than 20 reviews**:

```
1 airbnb[, gem := as.integer(star_rating >= 4.5 & reviews_count > 20)]  
2 table(airbnb$gem)
```

```
      0      1  
40460 10358
```

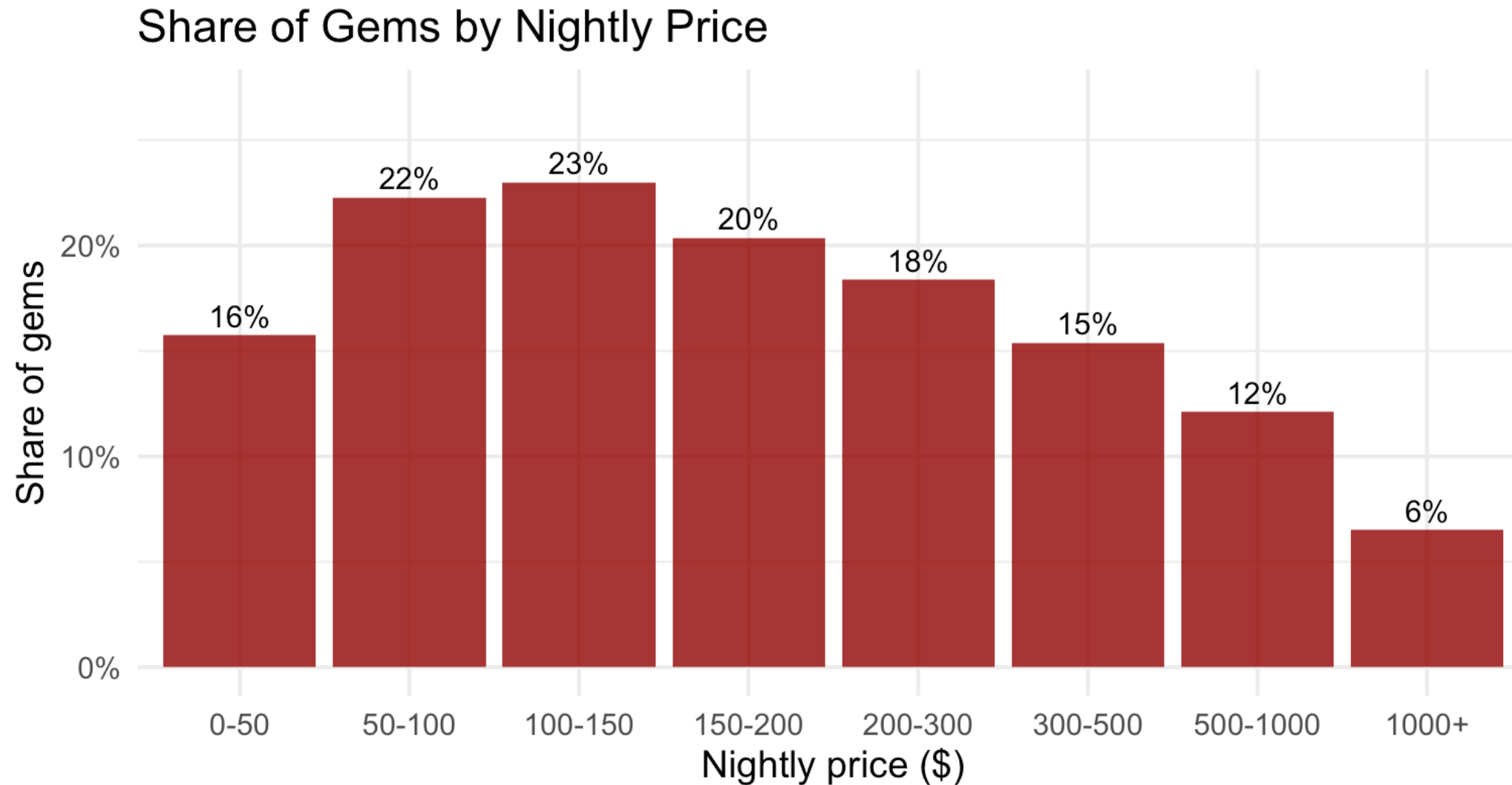
```
1 mean(airbnb$gem)
```

```
[1] 0.2038254
```

- **gem** is a **dummy**: 1 for a gem, 0 otherwise. About **20%** of listings are gems
- The question: **what makes a listing more likely to be a gem?** Start with price

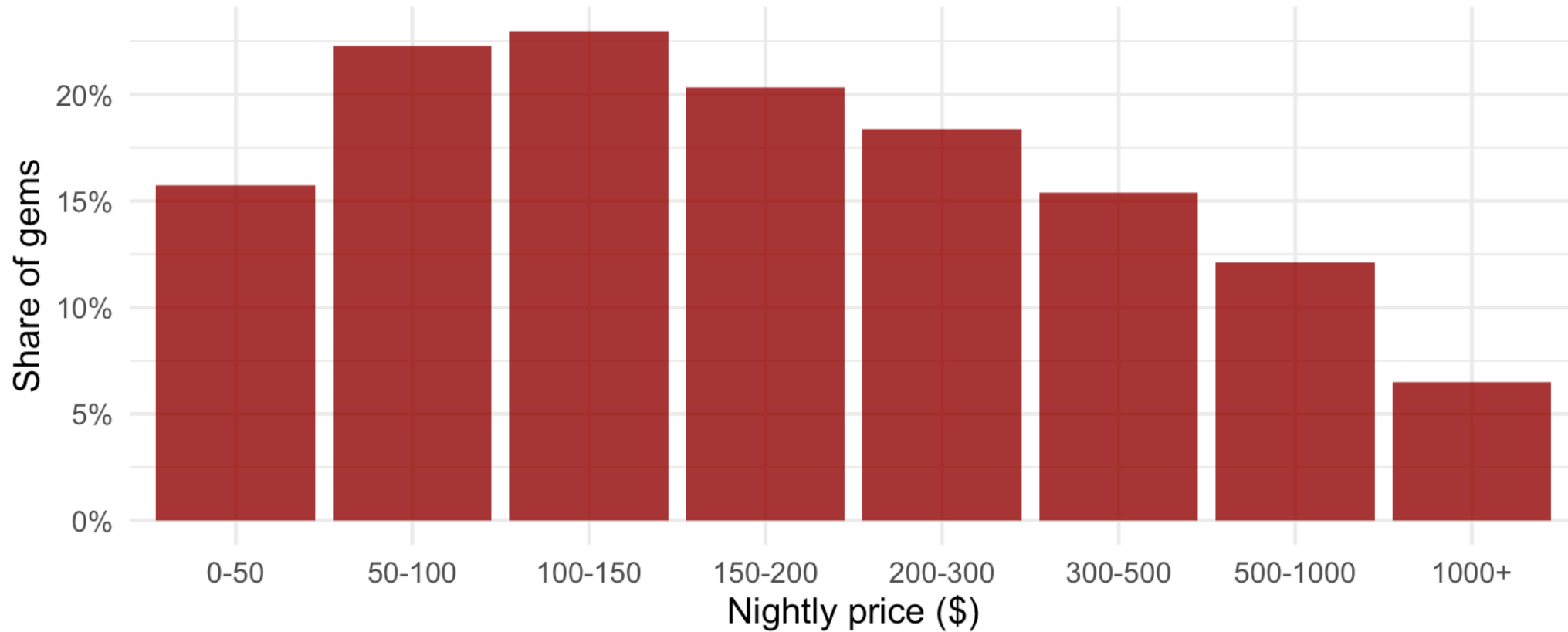
The **&** means both conditions must hold; **as.integer()** turns TRUE/FALSE into 1/0, as in Task 7 of the case.

Share of gems by price



Same idea as the “average by group” charts of week 3, with a 0/1 variable: the average of a dummy is a **share**. What do you see?

Share of gems by price: reading it

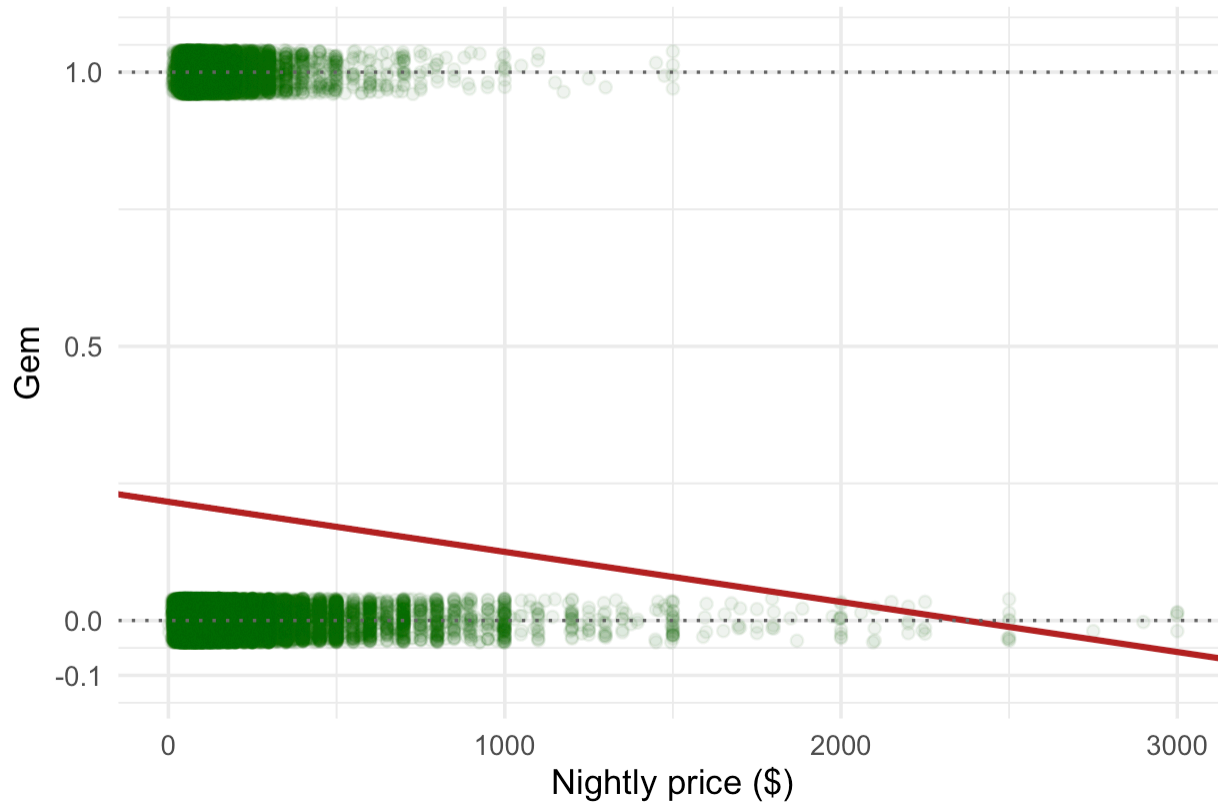


- Gems are **rarer as price rises**: 22–23% of listings between \$50 and \$150, 12% between \$500 and \$1,000, 6% above \$1,000. The very cheapest bin is also lower
- So the **probability** of being a gem depends on price. Can we model that probability with a regression, the way we modeled price on Thursday?

Try the tool we have: a straight line

```
1 lpm <- lm(gem ~ price, data = airbnb) # OLS, with a 0/1 outcome
```

Gem (0/1) vs. Price, with the OLS line



- The “linear probability model”: the line is the **predicted probability** of being a gem
- Intercept **0.217**: a free listing has a 21.7% chance. Slope **-0.00009**: each extra dollar lowers it by 0.009 percentage points; \$100 more, by about **0.9 points**
- Reasonable so far. Now look at the right end of the line

Where the line breaks

```
1 pred <- predict(lpm)
2 sum(pred < 0)           # listings with a predicted probability below zero
```

```
[1] 22
```

```
1 round(min(pred), 2)     # the lowest one
```

```
[1] -0.7
```

- The line crosses zero at a price of about **\$2,372** and keeps going: a \$10,000 listing gets a “probability” of **-70%**
- A probability is trapped **between 0 and 1**. As it approaches either edge, the true relationship must **flatten**. A straight line cannot flatten: it falls at the same rate forever
- The slope has the same flaw: the line takes **0.9 points** off for every \$100, whether a listing’s chance of being a gem is at 20% (a cheap listing) or at 2% (a very expensive one). At 2% there are only 2 points left to lose, so near the edge the effect of price must shrink

For a yes/no outcome we need a curve that is **linear in the middle** and **flattens at both ends**. That curve is the logistic function, and the model built on it is the **logit** (logistic regression).

Logistic Regression

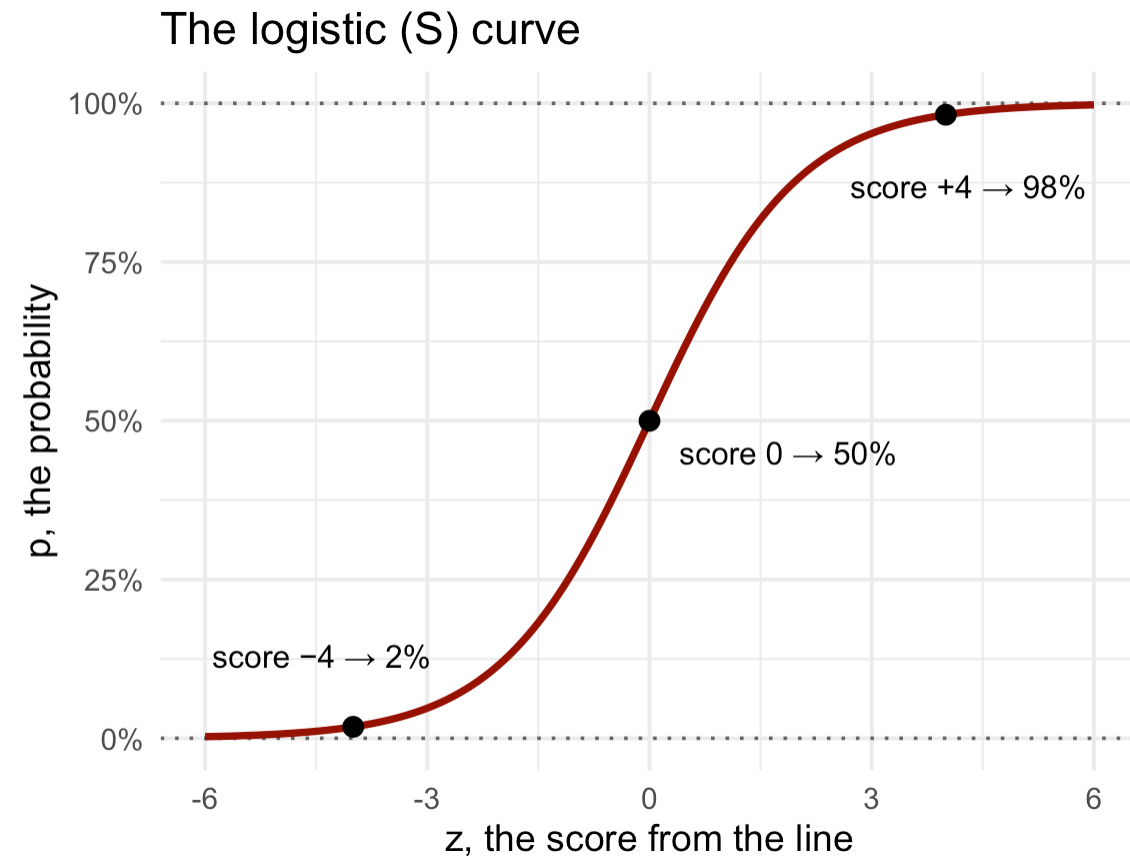
The fix: squash the line into 0–1

Two steps, and only the second one is new:

1. Keep Thursday's straight line, but treat its output as a **score**, not as the probability:
 $z = \beta_0 + \beta_1 X$. A score can be any number:
0.3, 12, -70

2. **Squash** the score into 0–1 with the **logistic function**: $p = \frac{1}{1 + e^{-z}}$. Any score in, a probability out

- A huge negative score becomes a probability near 0, **never below it**; a huge positive one, near 1, never above
- You will never compute this by hand: R does. Your job is to read what comes out



Odds and log-odds

The **odds** are the chance it happens, divided by the chance it does not. Sports betting already speaks this language:

Probability	Odds = $p/(1 - p)$	In words
$p = 20\%$ (the chance a random listing is a gem)	$0.2/0.8 = 0.25$	one to four: 1 gem for every 4 non-gems
$p = 50\%$	$0.5/0.5 = 1$	even: as likely as not
$p = 80\%$	$0.8/0.2 = 4$	four to one: 4 gems for every non-gem

- Rearranging the S-curve puts the model on this scale: $\frac{p}{1 - p} = e^z$, and taking logs,
$$\log\left(\frac{p}{1 - p}\right) = z = \beta_0 + \beta_1 X$$
- The right-hand side is Thursday's **straight line**. That is what “logit” means, and why everything from OLS carries over: predictors, dummies, base levels, standard errors, stars

A logit coefficient β_1 is the change in the **log-odds** for one more unit of X . Nobody thinks in log-odds, so the next slide converts.

Reading a logit coefficient

The rule: one more unit of X **multiplies the odds by** e^{β_1} ; R computes it for you (`exp(coef(m))`). The worked example previews the model we run next, a logit of **gem on price**: its coefficient will come out at $\beta_1 = -0.0008$ per dollar. Your job is three steps:

Step	Price example: $\beta_1 = -0.0008$ per dollar
1. Sign and stars , as in OLS	negative, significant: pricier listings are less likely to be gems
2. Scale : multiply β_1 by the change you care about	for \$100 more: $100 \times -0.0008 = -0.08$
3. Convert : odds are multiplied by e^{β_1} ; the percent change is $e^{\beta_1} - 1$	$e^{-0.08} = 0.923$: \$100 more multiplies the odds by 0.923, and $0.923 - 1 = -0.077$: a 7.7% drop in the odds

- Shortcut, as in Thursday’s log-level model: while the scaled coefficient is within about ± 0.2 , just move the decimal ($-0.08 \rightarrow$ about -8%); bigger than that, use `exp()`

The 8% drop is a change in the **odds**, not in the probability of being a gem. What it does to the probability takes one more conversion: next slide.

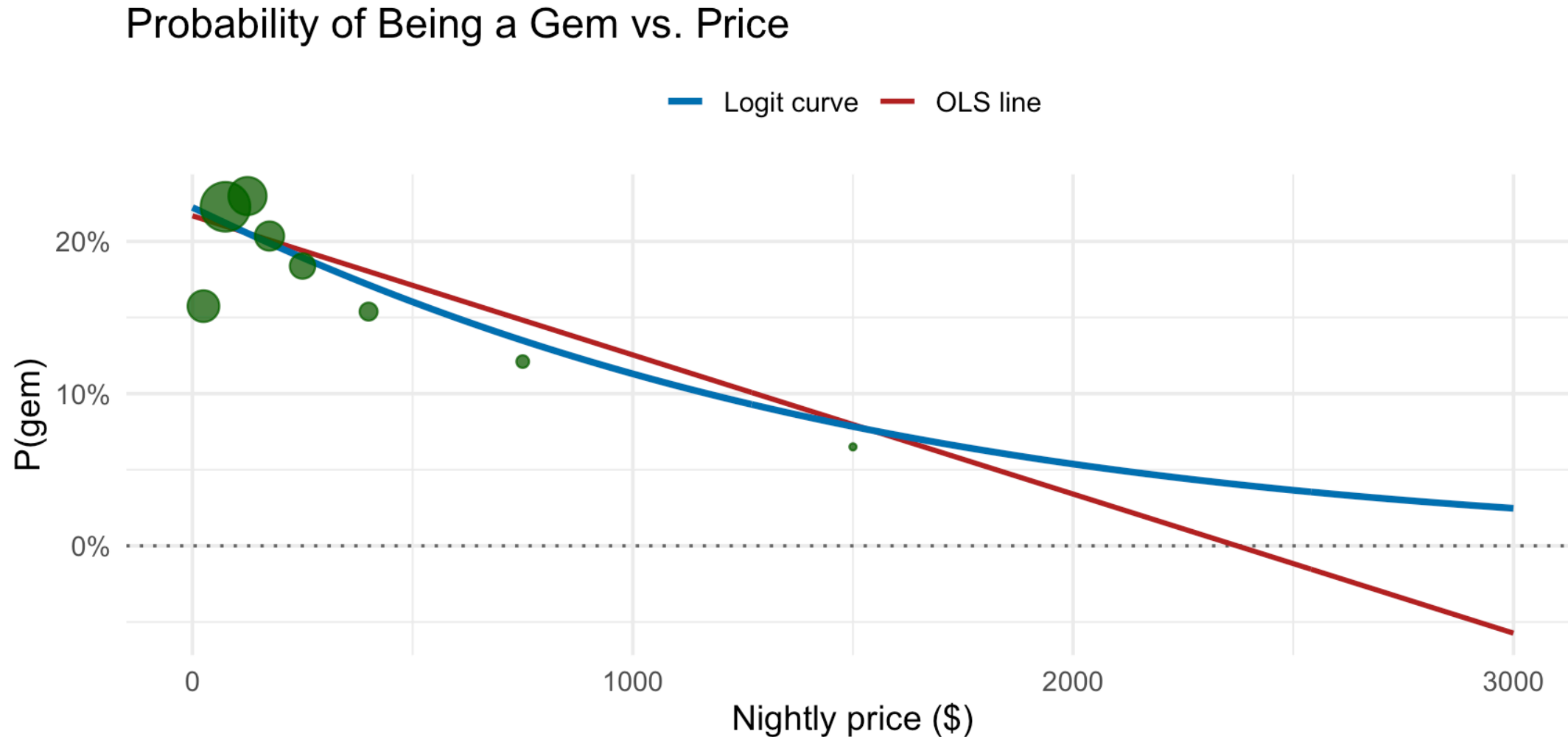
Odds fall 8%: what happens to the probability?

Not “minus 8 points”. The model speaks on the **odds** scale, so walk the change through the odds and back, one step at a time:

1. **Write the probability as odds.** A typical listing has a **20%** chance of being a gem. As odds: $0.2/0.8 = 0.25$ (slide 14's table, first row)
2. **Apply the change to the odds**, because that is where it lives. \$100 more multiplies the odds by 0.923, so:
 $0.25 \times 0.923 = 0.231$
3. **Turn the new odds back into a probability**, with the conversion formula: $p = \frac{\text{odds}}{1 + \text{odds}} = \frac{0.231}{1.231} = 18.8\%$

So \$100 more takes the chance of being a gem from 20% to **18.8%**: a drop of **1.2 percentage points**, not 8. The “8%” lives on the odds scale; the probability always moves by less, and the gap between the two scales grows with the size of the change.

Logit on price: the curve, on the data



Green dots: the observed share of gems in each price bin (sized by number of listings). Which model fits the dots, and what happens past \$2,000?

The curve: reading it

```
1 lg1 <- glm(gem ~ price, data = airbnb,  
2           family = binomial)  
3 stargazer(lg1, type = "text",  
4           omit.stat = c("f", "ser", "aic", "bic"),  
5           digits = 4, no.space = TRUE)
```

Dependent variable:	
Gem (0/1)	
Price	-0.0008*** (0.0001)
Constant	-1.2520*** (0.0163)
Observations	50,818
Log Likelihood	-25,650

- Every regression you report goes in a **table**, logit included: same `stargazer` call as Thursday
- The curve never dips below zero: at \$2,372, where the line hits zero, it still says about **4%**; at \$10,000, essentially zero
- The coefficient, with the recipe. **Sign and stars:** negative, significant. **Scale**, for \$100:
 $100 \times -0.0008 = -0.08$
- **Convert:** the shortcut says about an **8% drop in the odds** per \$100; the exact number, $e^{-0.08} - 1 = -7.7\%$, agrees

Odds are for reading the table. What a manager wants is the **predicted probability**: the chance that a listing at a given price is a gem. Next slide: how to get it, at any price you like.

A probability at any price

The two steps from “the fix” slide, run forward with the two numbers in the table: **score, then squash**.

1. **Score**. Take the constant and the price coefficient from the table. At \$500:

$$z = -1.252 + (-0.0008 \times 500) = -1.65$$

2. **Squash**. $p = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{1.65}} = 0.16$: a \$500 listing has a **16%** chance of being a gem

`predict()` runs both steps for you. Give it the **saved model** (`lg1`, which remembers the two coefficients), the prices you want as a little table, and `type = "response"`, which means “answer in probabilities”:

```
1 prices <- data.frame(price = c(50, 500, 5000))
2 round(predict(lg1, prices, type = "response"), 3)
```

```
   1     2     3
0.215 0.160 0.005
```

One probability per price: **21.5%** at \$50, **16%** at \$500 (matching the hand computation above), **0.5%** at \$5,000. The blue curve is exactly these numbers, computed at every price.

How it is estimated, and how to judge the fit

Estimation. OLS picks the line with the smallest squared errors. Logit picks the coefficients that make the observed 0s and 1s **most likely** (**maximum likelihood**). R does it in a fraction of a second; you never need the formula.

Log-likelihood. How likely the data is under the model. Always negative; **closer to zero is better**. Use it to compare models on the same data, the way you used R^2 on Thursday.

For the price-only model: log-likelihood -25,650 vs. -25,697 without price; pseudo- R^2 0.002. Price matters, and explains very little, the same lesson as Thursday.

There is no R^2 . Alternatives:

- **Pseudo- R^2** (McFadden): how much the log-likelihood improves over a model with no predictors. 0 to 1, but values of 0.1–0.3 are already decent
- **Accuracy:** predict “gem” when $p > 0.5$ and count the hits. Beware the base rate: saying “not a gem” for **everyone** is right **80%** of the time here. Week 9 (classifiers) covers this properly

Logit in R

One function: glm()

glm() works like lm(), with family = binomial to ask for a logit:

```
1 m_logit <- glm(gem ~ price + guests_included + city + room_type, data = airbnb, family = binomial)
2 stargazer(m_logit, type = "text", omit.stat = c("f", "ser", "aic", "bic"),
3           digits = 4, no.space = TRUE)
```

	Dependent variable:
	Gem (0/1)
Price	-0.0017*** (0.0001)
Guests included	0.1119*** (0.0079)
Boston	0.1674*** (0.0405)
Los Angeles	0.1617*** (0.0293)
Miami	-0.1955*** (0.0416)
Private room	-0.0072 (0.0265)
Shared room	-0.7068*** (0.0642)
Constant	-1.3935*** (0.0349)
Observations	50,818
Log Likelihood	-25,420

Same table, typeset for the slide; in your terminal it prints as text. Standard errors in parentheses under each coefficient; stars: *p < 0.1, **p < 0.05, ***p < 0.01. Read the signs, then the sizes. Which predictor has the largest effect?

Reading the table

Coefficient	Odds multiply by	Reading (holding the others fixed)
Price −0.0017	$e^{-0.17} = 0.84$ per \$100	\$100 more: odds of being a gem 16% lower
Guests included 0.11	1.12 per guest	one more guest included: odds 12% higher
Boston 0.17, Los Angeles 0.16	1.18, 1.18	vs. Austin: odds about 18% higher
Miami −0.20	0.82	vs. Austin: odds 18% lower
Private room −0.01, not significant	0.99	same odds as an entire home
Shared room −0.71	0.49	odds halved vs. an entire home

- The **largest** effect is the shared-room dummy, not price: a \$100 price difference moves the odds less than the room type does
- Log-likelihood −25,420, vs. −25,697 with no predictors

From odds to probabilities

Odds ratios compare. Probabilities decide. `predict()` with `type = "response"` gives the probability for any listing you describe:

```
1 examples <- data.frame(price = c(100, 100, 300, 100), guests_included = 2,  
2                           city = c("Austin", "Boston", "Austin", "Austin"),  
3                           room_type = c("Entire home", "Entire home", "Entire home", "Shared room"))  
4 examples$p_gem <- round(predict(m_logit, examples, type = "response"), 3)  
5 examples
```

	price	guests_included	city	room_type	p_gem
1	100	2	Austin	Entire home	0.208
2	100	2	Boston	Entire home	0.236
3	300	2	Austin	Entire home	0.157
4	100	2	Austin	Shared room	0.114

- An entire home in Austin at \$100 for 2 guests: **21%** chance of being a gem. The same listing in Boston: **24%**. At \$300 instead of \$100: **16%**. As a shared room: **11%**
- This is the sentence for the investor: “moving the same property from Austin to Boston raises the chance it becomes a gem from 21% to 24%”

Odds ratios for a table, probabilities for a decision. Always compute a few predicted probabilities for realistic cases before you write a recommendation.

OLS and logit, side by side

	(Thursday)	(today)
Outcome	a number (price, sales)	yes/no (gem, buy, churn, click)
Model	$Y = \beta_0 + \beta_1 X$	$\log \frac{p}{1-p} = \beta_0 + \beta_1 X$
R	<code>lm(y ~ x)</code>	<code>glm(y ~ x, family = binomial)</code>
β_1 means	one more unit of $X \rightarrow \beta_1$ more units of Y	one more unit of $X \rightarrow \text{odds} \times e^{\beta_1}$
Predictions	<code>predict(m)</code>	<code>predict(m, type = "response")</code> gives probabilities
Fit	R^2 , residual standard error	log-likelihood, pseudo- R^2
Estimation	least squares	maximum likelihood
Same in both	predictors, logs of X , dummies and base levels, SE, stars, “holding the others fixed”, not causation	

Still not causation

Cheaper listings are more likely to be gems. Three stories, as always:

1. **Price causes gem status**: lower price → more bookings → more reviews → gem
2. **Gem status causes price**: a host whose listing sits empty cuts the price; a host booked solid at 4.8 stars does not have to. The price reacts to success, it does not create it
3. **Something else causes both**: gems are **older** (20+ reviews take time), and older listings were priced in an earlier, cheaper market

“A \$100 higher price is **associated with** 16% lower odds of being a gem.” The model holds fixed guests, city, and room type. It does not hold fixed listing age, photos, or host effort.

The Airbnb Case

The case

The handout ([w4-airbnb-case.html](#)) works like RateBeer, with less help: an exact prompt for Task 1, skeletons for Tasks 2–3, and only the goal from Task 4 on.

The client is an **investor** deciding where to buy and what to list:

1. **Tasks 1–3**: look at the data, price vs. reviews with and without logs, the three models side by side (Thursday, applied)
2. **Tasks 4–5**: more predictors, then city as a factor and as hand-made dummies
3. **Tasks 6–7**: a revenue proxy, room type, a quality dummy
4. **Task 8**: three recommendations, one of them a limitation

Under every task: **predict** before you run, **explain back** one line after.

Aim to finish **Task 5** in class. Deliverable, same as always: an **R Markdown report knitted to HTML** plus [ai-log.md](#), due **Sunday, Sept. 27, 11:59 pm**. Data: [airbnb-case-data.zip](#).

Wrap-up

Code and data to reproduce today's models, on the course website:

- [w4-code.zip](#): both week 4 scripts and the data
- [w4-2-logit-class.R](#): every chart and model shown today
- [The Airbnb case handout](#) and its [data](#)

Required reading: Chapter [3.6](#) of *R for Marketing Students*. Optional: Chapter 9.2 of *R for Marketing Research and Analytics*.

Questions?